WoSM Academy

CS102-F26 · DIGITAL DESIGN AND COMPUTER ARCHITECTURE I

# Course Review

## Lecture 23: from switches to a computer, and the exam

Dr. Ali Khudiyev

Fall 2026



CC BY-SA 4.0

# Today

01 The whole journey

02 Review by learning outcome

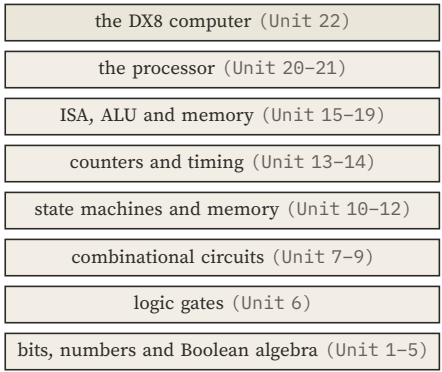
03 The exam

04 Where next

01

# The whole journey

# From switches to a computer




- Each layer is built only from the one below it.
- The first half of the module: the foundations of digital design. The second half: putting them together into a working 8-bit computer.
- The six learning outcomes of the module follow the same order: the next slides take them one at a time.

02

# Review by learning outcome

# LO1: numbers

You should be able to

- tell additive from positional numeral systems;
- convert between decimal, binary, octal and hexadecimal;
- group binary digits in threes (octal) or fours (hexadecimal).

Units 1 and 2

### Practice

1. Write `0x2F` in binary and in decimal.
   0010 1111 = 47.
2. Write 156 in octal.
   $156 = 2 \cdot 64 + 3 \cdot 8 + 4$: $234_8$.

# LO2: Boolean algebra

You should be able to

- use the laws and theorems of Boolean algebra, De Morgan's among them;
- write the canonical sum of products and product of sums of a truth table;
- simplify a function of up to four variables with a Karnaugh map.

Units 3, 4 and 5

## Practice

1. Simplify $xy + xy' + x'y$.
   $x(y + y') + x'y = x + x'y = x + y$.
2. Simplify $f(x, y, z) = \sum m(1, 3, 5, 6, 7)$ with a K-map.
   $f = z + xy$.

# LO3: combinational circuits

You should be able to

- build any function from a universal gate set, NAND alone included;
- design multiplexers, decoders, encoders, adders and comparators;
- find propagation and contamination delays, and spot glitches.

Units 6 to 9, 16 and 17

## Practice

1. Build $x + y$ from NAND gates only.
   $x + y = (x' \, y')'$: NAND the outputs of $\text{NAND}(x, x)$ and $\text{NAND}(y, y)$.
2. How many select bits does a 32-to-1 multiplexer need?
   5, since $2^5 = 32$.
3. What is $-100$ as an 8-bit two's complement number?
   1001 1100.

# LO4: sequential circuits

You should be able to

- model a problem as a Moore or a Mealy machine, and simplify it;
- explain the SR and D latches and the D, T and JK flip-flops;
- build registers and counters;
- apply the setup and hold time constraints.

Units 10 to 14

## Practice

1. A 4-bit up counter starts at 0. What does it show after 20 rising edges?
   $20 \bmod 16 = 4$: 0100.
2. $t_{\text{clk2q}} = 50$ ps, $t_{\text{pd}} = 400$ ps, $t_{\text{setup}} = 50$ ps. What is the fastest clock?
   $t_{\text{clk}} \geq 500$ ps: 2 GHz.

# LO5: architecture, ALU and memory

You should be able to

- explain the Von Neumann architecture and what an ISA is;
- encode and decode DX8 instructions;
- design an ALU, a register file and byte-addressable RAM from basic parts.

Units 15 to 19

**Practice**

1. Write $M[\texttt{0x33}] \leftarrow$ R6 (a `store`) in hexadecimal.
   0 0001 110, 0011 0011: `0x0E33`.
2. How many address bits does a byte-addressable RAM of 64 bytes need?
   6.

# LO6: the processor

You should be able to

- explain fetch, decode, execute and load;
- compare single-cycle, multi-cycle and pipelined designs;
- build the DX8 in a circuit simulator, and program it.

Units 20 to 22

**Practice**

1. 4 stages of 2 cycles each. How many cycles do 20 instructions take, multi-cycle and pipelined?
   160, and $2 \cdot (20 + 3) = 46$.
2. In which step does the DX8 copy the result of an `add` into a register?
   $s_7$, in the load stage.

03

# The exam

# How the module is assessed

| Component | Weight | Due | Outcomes |
|---|---|---|---|
| Quizzes | 20% | throughout | LO1 to LO5 |
| Circuit-design project: the DX8 computer | 40% | throughout | LO3 to LO6 |
| Final exam | 40% | end of module | LO1 to LO6 |

- The final exam covers every learning outcome: every unit of the module.
- Any support materials that are permitted will be provided during the exam. No other aids or resources are allowed.

# The final exam

- Date and time: `TODO(owner):` exam date and time
- Length: `TODO(owner):` length of the exam
- Format and kinds of questions: `TODO(owner):` exam format
- Support materials provided: `TODO(owner):` which materials are provided

> **How to prepare**
>
> Work through the exercise sheets and the “Try it” slides of every lecture again, rebuild the key circuits in Logisim-evolution, and practise encoding and tracing DX8 programs by hand.

04

# Where next

# Where next

- You have built a computer from switches that are on or off, one layer of abstraction at a time.
- *Digital Design and Computer Architecture II* (CS221-F27, Semester 3) continues with real instruction set architectures, memory systems, input and output, and processor performance.
- Keep your DX8: extend it, program it, and check its timing.

> **Remark**
>
> The book's epilogue says that it kept many explanations short, even hand-wavy, to stay under 100 pages, and it asks sharp readers for feedback. Finding those gaps yourself is the next step in learning.